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What is the False Discovery Rate in Finance?

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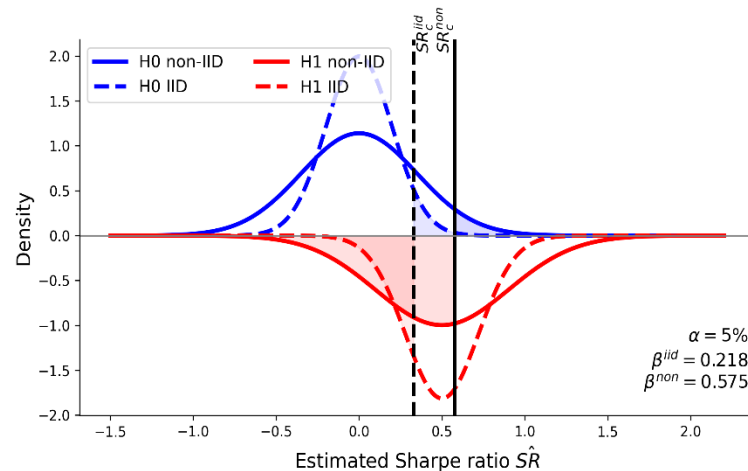
Seminar's Objective

- For the past 2 decades, Financial Economics has debated two opposing views
- One set of authors has raised concerns about
 - the lack of multiple testing corrections: claims are assessed as if they were the outcome of a single statistical experiment
 - poor performance post-publication: investors are not obtaining the returns that were promised
- Another set of authors has argued that there is no reproducibility crisis:
 - Using the cross-section of reported in-sample statistics, they estimate that the False Discovery Rate (FDR) in Finance is between 5% and 15%
- **In this seminar, we aim to resolve this apparent contradiction by showing that:**
 - The FDR is not identifiable from the cross-section of reported in-sample statistics, unless search and selection are explicitly modelled
 - Under mild search and selection, the reported cross-section is consistent with $FDR > 80\%$
- For additional details, find the full paper at https://ssrn.com/abstract_id=6450418

The Problem

Choice of Primitive Object

- Recall that t -statistics are asymptotically pivotal under the null only when the sampling variance of the mean is correctly specified
 - This typically requires i.i.d. Normal returns, or
 - asymptotics with a consistent variance estimator
- Consequently, any FDR analysis conducted in t -space inherits those assumptions
- However, it is well documented that investment strategy returns do not satisfy these assumptions
- To overcome this pitfall, we define false discovery probabilities in **Sharpe ratio space**, allowing for non-Normal and serially-correlated returns



The Generalized Variance of the Sharpe ratio enables a more accurate estimation of Type I and Type II errors, see [López de Prado et al. \[2026\]](#).

In the example above, incorrectly assuming i.i.d. Normal returns leads to underestimating beta by 45%.

The Single-Trial Assumption

- A common approach in the literature is to estimate the null proportion (π_0) from a mixture with components F_0 (under the null) and F_1 (under the alternative)

$$\widehat{SR} \sim \pi_0 F_0 + (1 - \pi_0) F_1$$

$$F[x] = P[\widehat{SR} < x] = \pi_0 F_0[x] + (1 - \pi_0) F_1[x]$$

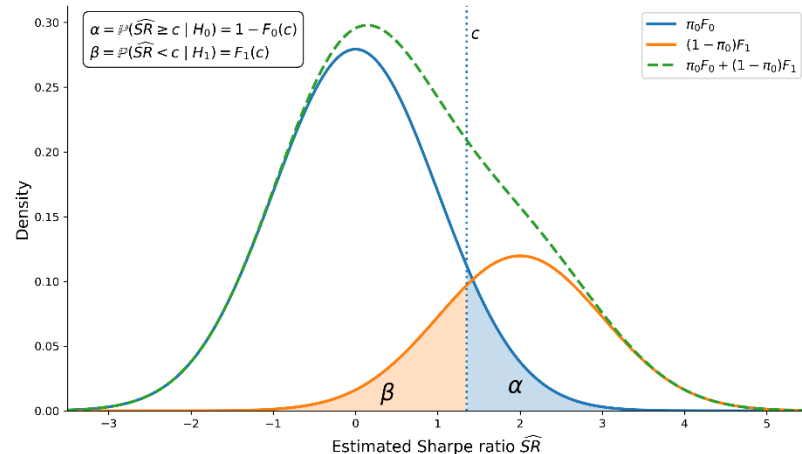
- Applying a rejection threshold c to F_0 and F_1 , Type I and II error probabilities and FDR are

$$\alpha = P[\widehat{SR} \geq c \mid H_0] = 1 - F_0[c]$$

$$\beta = P[\widehat{SR} < c \mid H_1] = F_1[c]$$

$$FDR[\alpha, \beta, \pi_0] = P[H_0 \mid \widehat{SR} \geq c]$$

$$= \frac{\alpha \pi_0}{\alpha \pi_0 + (1 - \beta)(1 - \pi_0)}$$

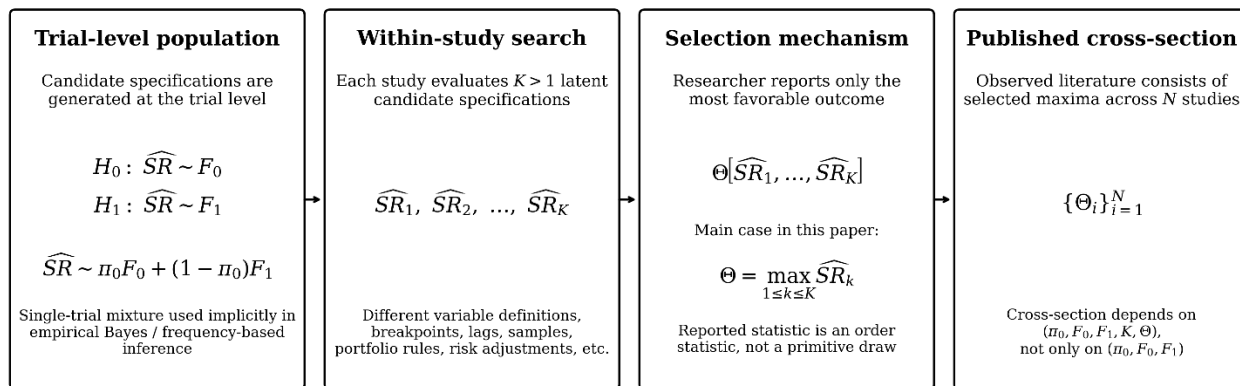


Most papers model the cross-section of published in-sample statistics using empirical Bayes / hierarchical Bayes / mixture-style models. These approaches assume that the “factor zoo” is a collection of anomalies, **where each anomaly is the result of one fixed specification that either enters or does not enter the published record.**

The Solution

A More Realistic Assumption

- Much of the empirical literature implicitly treats each reported statistic as the outcome of a single statistical trial drawn from the mixture $\pi_0 F_0 + (1 - \pi_0) F_1$
- In practice, however, a reported statistic is often generated by a latent search over K candidate specifications, followed by a selection rule $\Theta[\cdot]$ that retains only the most favorable outcome



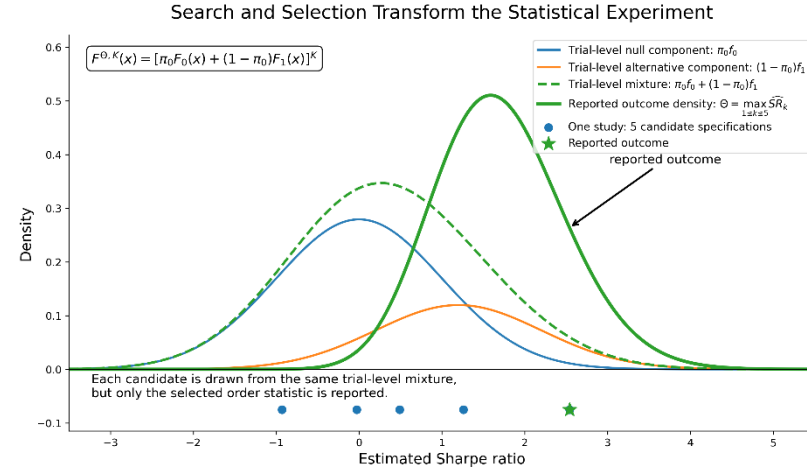
The “factor zoo” is not merely a collection of anomalies, where each anomaly is the result of **one fixed specification that either enters or does not enter the published record**. Rather, a recorded anomaly is often **the endpoint of a latent search over many alternative within-study specifications**.

$K = 1$ = standard single-trial mixture

$K > 1$ = selection-distorted experiment; FDR not identified without modeling (K, Θ)

Search and Selection (1/2)

- Under this more realistic framework, each candidate $\widehat{SR}_k^*$ is still drawn from the mixture,
$$\widehat{SR}_k \sim \pi_0 F_0 + (1 - \pi_0) F_1$$
- However, instead of reporting each $\widehat{SR}_k^*$, the researcher compares all K results, and among them reports only one according to some selection mechanism informed by the Sharpe ratios, $\Theta \left[\left\{ \widehat{SR}_k^* \right\}_{k=1}^K \right]$
- This selection mechanism changes the statistical experiment by replacing the single-trial outcome $\widehat{SR}^*$ with the selected order statistic $\Theta \left[\left\{ \widehat{SR}_k^* \right\}_{k=1}^K \right]$



Search and selection change the location and shape of the order-statistic distributions under the null and the alternative.

Search and Selection (2/2)

- The familywise probabilities are

$$\alpha_K = P \left[\Theta \left[\left\{ \widehat{SR}_k \right\}_{k=1}^K \right] \geq c \mid M = 0 \right] = 1 - F_0^{\Theta, K}[c]$$

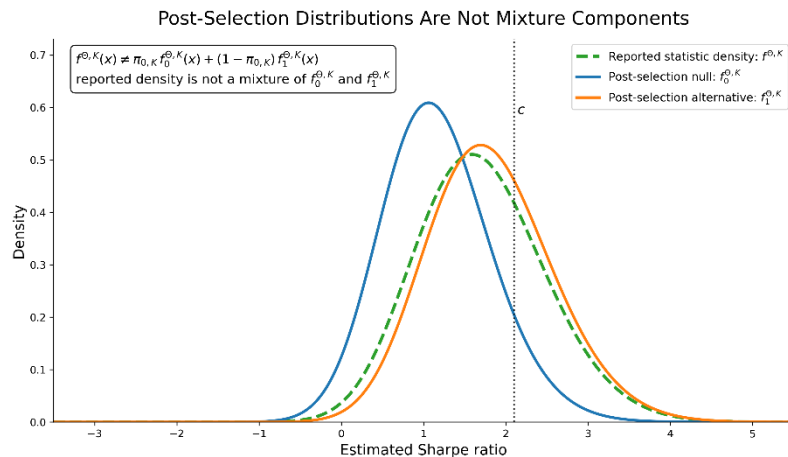
$$\beta_K = P \left[\Theta \left[\left\{ \widehat{SR}_k \right\}_{k=1}^K \right] < c \mid M \geq 1 \right] = F_1^{\Theta, K}[c]$$

$$\pi_{0,K} = P[M = 0] = \pi_0^K$$

where

- M is the number of non-null trials for a reported statistic, $M = \sum_{k=1}^K 1\{\widehat{SR}_k \sim F_1\}$, with $M \sim \text{Binomial}[K, 1 - \pi_0]$
- $F_0^{\Theta, K}[c]$ is the distribution under the null ($M = 0$)
- $F_1^{\Theta, K}[c]$ is the distribution under the alternative ($M \geq 1$)

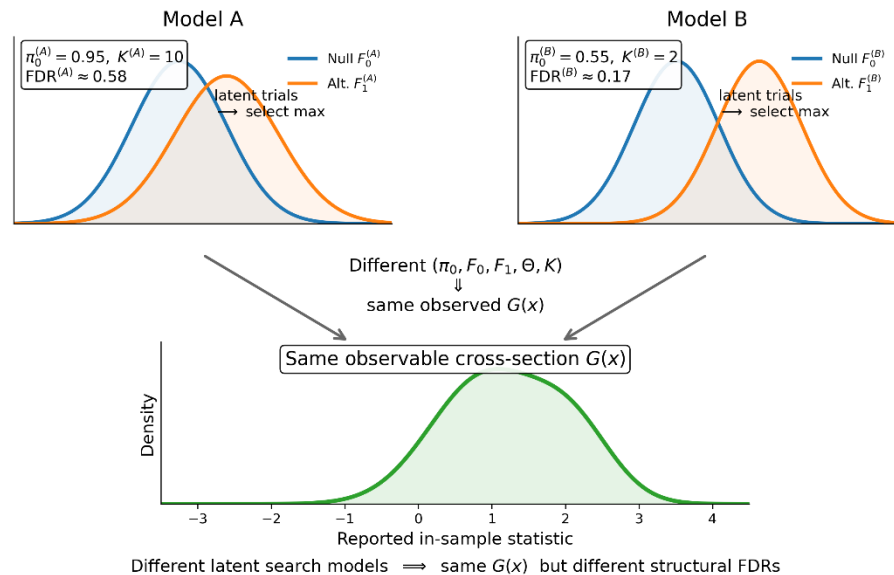
$$\begin{aligned} FDR[\alpha_K, \beta_K, \pi_{0,K}] &= P \left[H_0 \mid \Theta \left[\left\{ \widehat{SR}_k \right\}_{k=1}^K \right] \geq c \right] \\ &= \frac{\alpha_K \pi_{0,K}}{\alpha_K \pi_{0,K} + (1 - \beta_K)(1 - \pi_{0,K})} \end{aligned}$$



The distributions $F_0^{\Theta, K}$ and $F_1^{\Theta, K}$ are not components of a mixture, $\pi_{0,K} F_0^{\Theta, K} + (1 - \pi_{0,K}) F_1^{\Theta, K}$. A mixture model would be misspecified, and its estimates would be biased by (Θ, K) .

Identification Failure

- **Theorem 1 (Identification Failure):** Once reported results are generated by a latent search-and-selection process, the mapping from the structural tuple $(\pi_0, F_0, F_1, \Theta, K)$ to the observable cross-section G is not injective
- Implications
 - Different structural tuples can render the same observations
 - Empirical-Bayes, hierarchical-Bayes, local-FDR, and mixture-style methods applied directly to the selected cross-section cannot recover the structural FDR unless search and selection are explicitly modeled



Two structurally distinct search models can generate the same observable cross-section of reported statistics. This non-injectivity is the source of Theorem 1: the structural search-adjusted FDR is not identified from the selected cross-section alone.

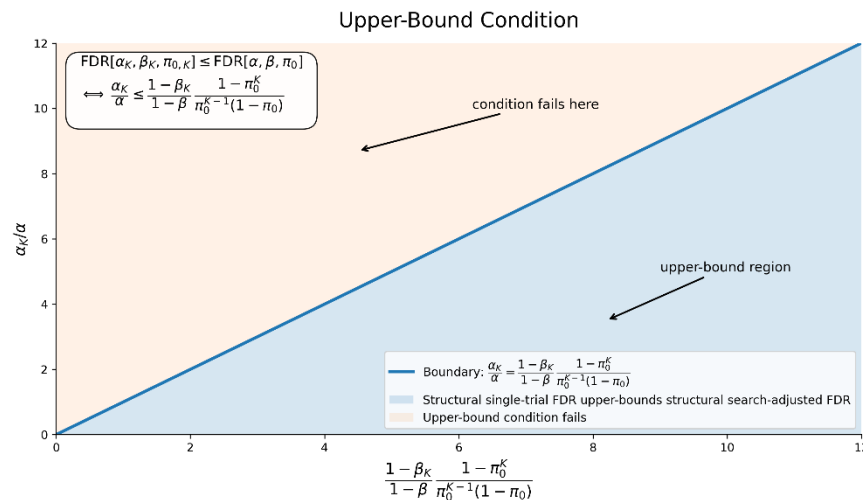
Upper Bound Condition

- We have seen that a mixture model fitted on the cross-section will result in biased estimates, but in what direction?
- Given $\alpha, \alpha_K, \beta, \beta_K, \pi_0, \pi_{0,K} \in (0,1)$, the upper bound

$$FDR[\alpha_K, \beta_K, \pi_{0,K}] \leq FDR[\alpha, \beta, \pi_0]$$

holds if and only if

$$\frac{\alpha_K}{\alpha} \leq \frac{1 - \beta_K}{1 - \beta} \cdot \frac{1 - \pi_0^K}{\pi_0^{K-1}(1 - \pi_0)}$$



The FDR computed under the unrealistic single-trial assumption can still work as an upper bound if and only if the inflation in false positive odds is not too large relative to both power distortion and the much lower familywise null prior. See the paper for a proof (Propositions 1 & 2).

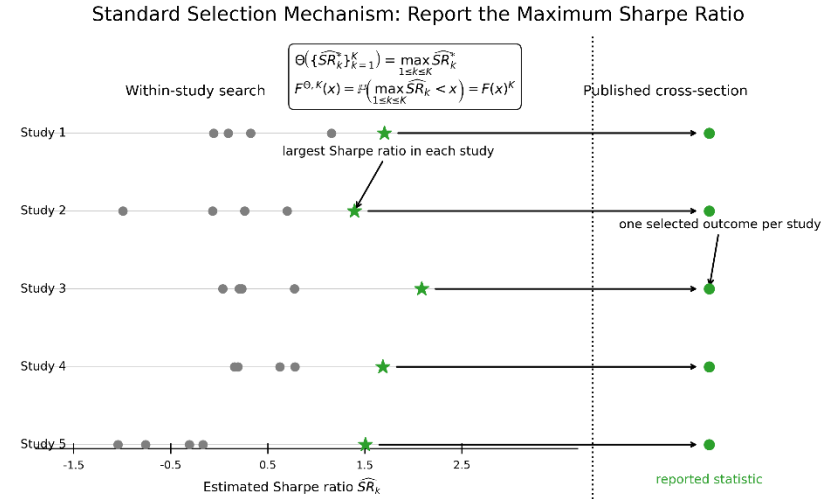
Example of Selection Mechanism (1/2)

- Consider the standard selection mechanism that chooses the strategy with highest Sharpe ratio among the K alternatives explored, thus

$$\Theta \left[\left\{ \widehat{SR}_k^* \right\}_{k=1}^K \right] = \max_{1 \leq k \leq K} \{ \widehat{SR}_k^* \}$$

- The distribution of the reported statistic is thus the distribution of the maximum of K independent draws from the mixture $\pi_0 F_0 + (1 - \pi_0) F_1$, with CDF

$$\begin{aligned} F^{\Theta, K}[x] &= P \left[\max_{1 \leq k \leq K} \{ \widehat{SR}_k \} < x \right] \\ &= \left[\pi_0 F_0[x] + (1 - \pi_0) F_1[x] \right]^K \end{aligned}$$



Within each study, the researcher evaluates K candidate specifications and reports only the one with the highest estimated Sharpe ratio. Under this mechanism, the published record is formed by selected maxima rather than by threshold-filtered single-trial draws.

Example of Selection Mechanism (2/2)

- The explicit modelling of (Θ, K) also allows us to derive the familywise error probabilities in terms of the trial-level error probabilities

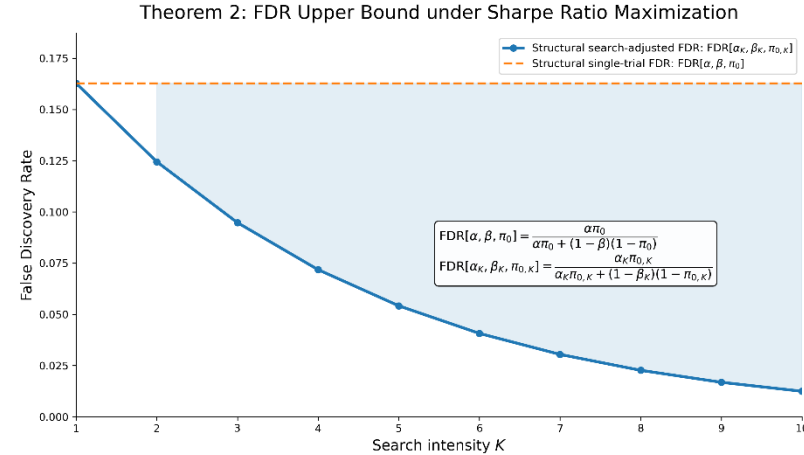
$$\alpha_K = P \left[\max_{1 \leq k \leq K} \{\widehat{SR}_k\} \geq c \mid M = 0 \right] = 1 - (1 - \alpha)^K$$

$$\beta_K = P \left[\max_{1 \leq k \leq K} \{\widehat{SR}_k\} < c \mid M \geq 1 \right]$$

$$= \frac{[\pi_0(1 - \alpha) + (1 - \pi_0)\beta]^K - [\pi_0(1 - \alpha)]^K}{1 - \pi_0^K}$$

- Theorem 2 (FDR Upper Bound under Sharpe Ratio Maximization):** If $K > 1$ and maximum Sharpe ratio is selected, then

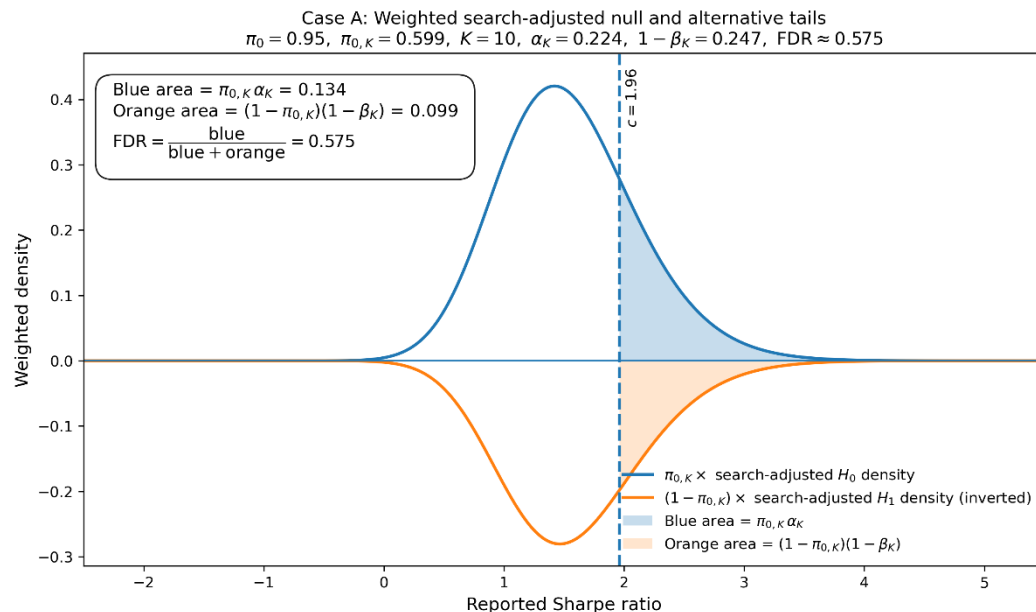
$$FDR[\alpha_K, \beta_K, \pi_{0,K}] < FDR[\alpha, \beta, \pi_0]$$



Theorem 2 is structural, not operational: under Sharpe ratio maximization, the structural single-trial FDR upper-bounds the structural search-adjusted FDR. But by Theorem 1, this bound cannot be recovered empirically from the selected cross-section alone, because latent search breaks identification.

Numerical Example A: FDR with Search & Selection

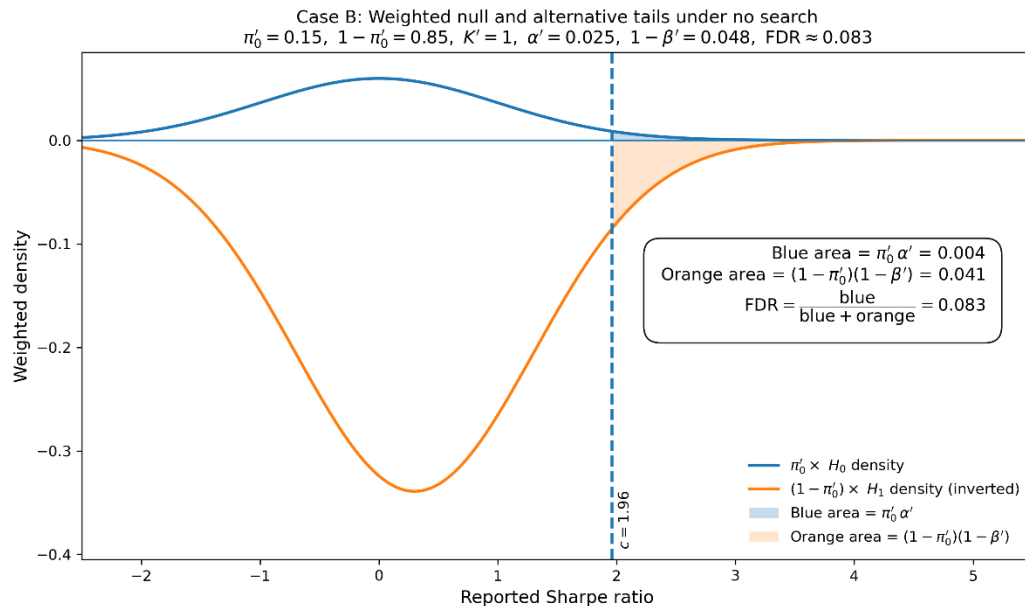
- Suppose the null distribution $H_0: \widehat{SR} \sim \mathcal{N}[0,1]$ with prevalence $\pi_0 = 0.95$, and the alternative distribution $H_1: \widehat{SR} \sim \mathcal{N}[\delta_1, 1]$ with mean $\delta_1 = 0.3$
- At the conventional rejection threshold $c = 1.96$, the error probabilities at $K = 10$ are
 $\alpha = 1 - Z[c] \approx 0.025$; $\alpha_K \approx 0.224$
 $\beta = Z[c - \delta_1] \approx 0.952$; $\beta_K \approx 0.753$
 $\pi_{0,K} = \pi_0^K$; $\pi_{0,K} \approx 0.599$
- The search-adjusted FDR is then
 $FDR[\alpha_K, \beta_K, \pi_{0,K}] \approx 0.575$



This figure plots the weighted search-adjusted null and alternative densities. The shaded right-tail areas are $\pi_{0,K} \alpha_K$ and $(1 - \pi_{0,K})(1 - \beta_K)$. **Once search and selection are present, the relevant objects are the post-selection densities rather than the single-trial distributions.**

Numerical Example B: FDR with No Search

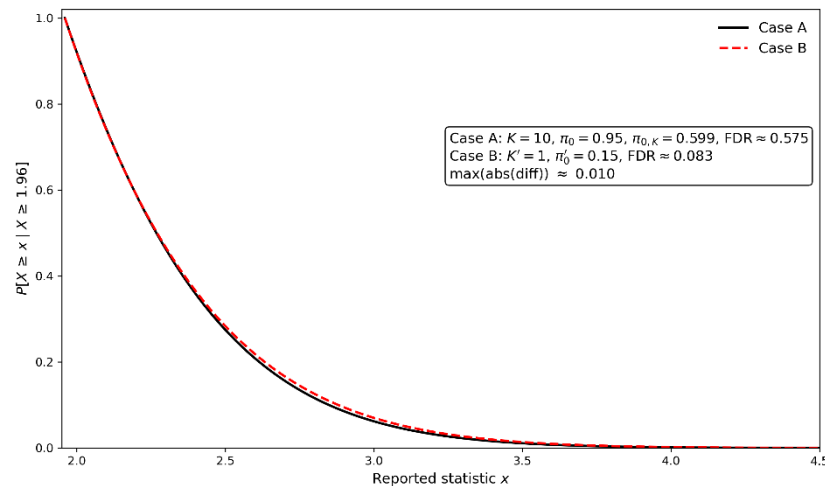
- Now consider an alternative data-generating process in which researchers report every trial ($K' = 1$), and prevalence of nulls $\pi'_0 = 0.15$
- Everything else remains the same
- The error probabilities are still
$$\alpha = 1 - Z[c] \approx 0.025$$
$$\beta = Z[c - \delta_1] \approx 0.952$$
- The no-search FDR is then
$$FDR[\alpha, \beta, \pi_0] \approx 0.083$$
- This follows the convention in the Financial Economics literature



This figure plots the weighted null and alternative densities. The shaded right-tail areas are $\alpha' \pi'_0$ and $(1 - \beta')(1 - \pi'_0)$. This figure provides the no-search comparison to the previous figure.

Numerical Example: Identification Failure

- **Question:** “Are these two data-generating processes distinguishable on the part of the distribution used for inference?”
- **Answer: No**, because
 - two distinct underlying discovery processes generate nearly identical conditional upper-tail behavior among reported results
 - the procedures that infer the prevalence of nulls or the FDR from the observed cross-section of reported statistics cannot reliably distinguish between them
- **Conclusion:** Once we accept that the single-trial assumption is unrealistic, earlier attempts to estimate the FDR from the cross-section of reported in-sample statistics fail!

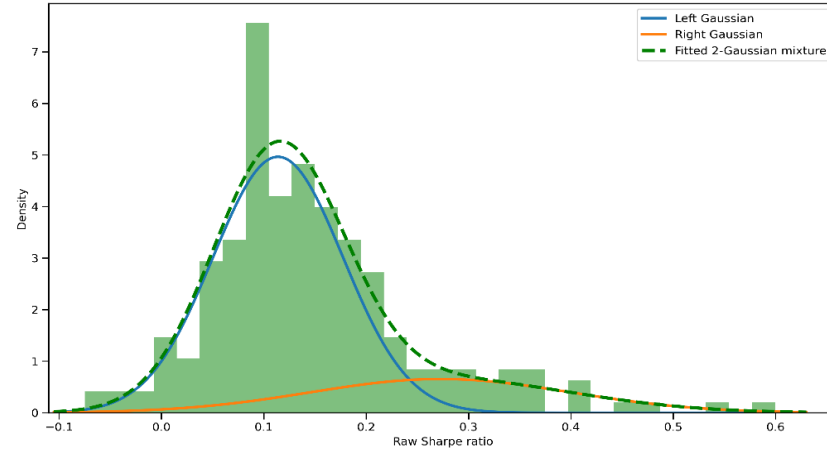


A high- π_0 world with moderate search and selection (Case A, $K = 10$) can produce reported significance profiles that are observationally equivalent, in the empirically relevant upper tail, to the profile generated by a low- π_0 world with no search (Case B, $K = 1$), even though the implied FDR values are radically different.

Empirical Calibration

The Data

- We use public data from the [Open Asset Pricing website](#) maintained by Andrew Chen and Tom Zimmermann
 - The file contains monthly long-short returns for 212 published predictors
 - The sample runs from January 1926 to December 2024, for a total of 1,188 monthly observations
 - The number of observations per predictor ranges from 197 to 1,188, with mean 817.46 and median 858
 - The mean first-order autocorrelation is 0.071 (median 0.075)
 - The empirical cross-section of predictor Sharpe ratios is positively skewed, leptokurtic, and concentrated almost entirely on positive values (fewer than 4% of values are negative)



If observations were filtered outcomes of single-trial experiments, we would expect the empirical distribution to exhibit features of truncation, such as an absence of negative values and mass accumulation at a threshold. Instead of truncation, the empirical distribution includes negative values, and is tilted toward the upper tail of the underlying sampling distribution. **These features align better with a search-and-selection process than with filtered single trials.**

Maximum of Mixtures Distribution

- Under search-and-selection, the CDF is

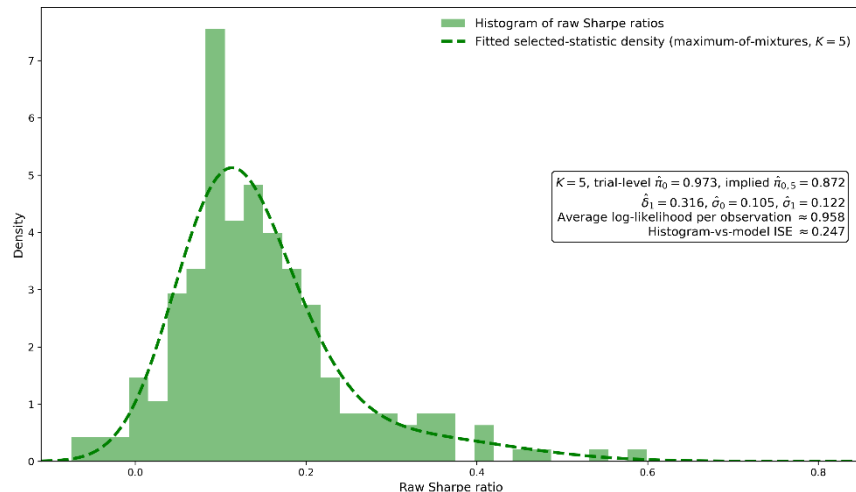
$$F^{\Theta,K}[x, \pi_0, \delta_1, \sigma] = P \left[\max_{1 \leq k \leq K} \{\widehat{SR}_{n,k}\} < x \right]$$

$$= [\pi_0 F_0[x, \sigma_0] + (1 - \pi_0) F_1[x, \delta_1, \sigma_1]]^K$$
- Differentiating the CDF, we obtain the PDF
 $f^{\Theta,K}[x, \pi_0, \sigma_0, \delta_1, \sigma_1]$, and the parameters
 are estimated by MLE as

$$(\hat{\pi}_0, \hat{\sigma}_0, \hat{\delta}_1, \hat{\sigma}_1)$$

$$= \arg \max_{\pi_0, \sigma_0, \delta_1, \sigma_1} \sum_{n=1}^N \log f^{\Theta,K}[\widehat{SR}_n, \pi_0, \sigma_0, \delta_1, \sigma_1]$$

- To avoid economically implausible local optima, we impose $\delta_1 \geq 0.1$, $\pi_0 \in (0,1)$, $\sigma_0 > 0$, and $\sigma_1 > \sigma_0$



The density for the maximum of mixtures distribution fits well the cross-section of reported in-sample Sharpe ratios.

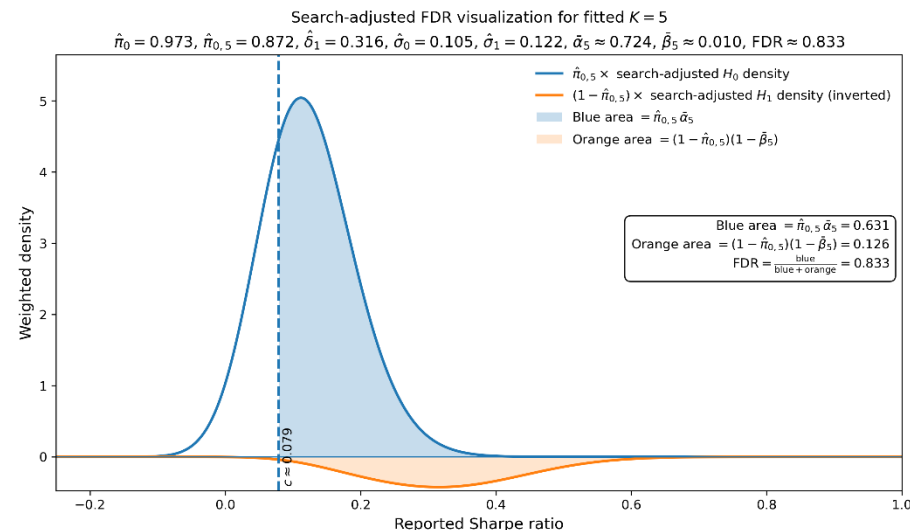
Search-Adjusted FDR (1/2)

K	pi_0	delta_1	sigma_0	sigma_1	pi_0_K	alpha_K	beta_K	log_likelihood	FDR
1	0.000	0.147	0.066	0.104	0.000	0.127	0.255	179.985	0.000
2	0.218	0.100	0.081	0.120	0.048	0.317	0.247	186.061	0.021
3	0.674	0.108	0.096	0.142	0.306	0.507	0.216	194.418	0.222
4	0.900	0.155	0.105	0.169	0.656	0.643	0.138	201.685	0.588
5	0.973	0.316	0.105	0.122	0.872	0.724	0.010	203.046	0.833
6	0.978	0.325	0.100	0.114	0.874	0.764	0.005	199.413	0.841
7	0.980	0.321	0.095	0.113	0.865	0.792	0.005	192.175	0.836
8	0.981	0.315	0.091	0.113	0.855	0.813	0.005	182.483	0.828
9	0.981	0.308	0.087	0.113	0.845	0.829	0.005	171.046	0.819
10	0.945	0.100	0.081	0.202	0.566	0.823	0.087	158.464	0.541

- The no-search and very-low-search cases fit the data poorly:** for $K = 1$ and $K = 2$, the estimated null prevalence remains low, with $FDR \approx 0.000$ and $FDR \approx 0.021$, respectively
- Once modest search is allowed, the fit improves sharply and the estimated null prevalence rises substantially:** at $K = 3$, the calibration yields $\hat{\pi}_0 \approx 0.674$, $\hat{\delta}_1 \approx 0.108$, $\hat{\sigma}_0 \approx 0.096$, $\hat{\sigma}_1 \approx 0.142$, $\hat{\pi}_{0,3} \approx 0.306$, and $FDR \approx 0.222$
- The log-likelihood is maximized at $K = 5$,** where $\hat{\pi}_0 \approx 0.973$, $\hat{\delta}_1 \approx 0.316$, $\hat{\sigma}_0 \approx 0.105$, $\hat{\sigma}_1 \approx 0.122$, $\hat{\pi}_{0,5} \approx 0.872$, $\bar{\alpha}_5 \approx 0.724$, $\bar{\beta}_5 \approx 0.010$, and $FDR \approx 0.833$. A similar FDR is obtained for 5 different search intensities

Search-Adjusted FDR (2/2)

- $\hat{K} \approx 5$ is the value that best fits the observed cross-section under our maintained parametric search model
- The model does not require that trials are independent
 - Search intensity is *expressed* in terms of the number of independent trials that would have the same effect on FDR as the actual number of dependent trials, hence the “effective” qualifier
- Accounting for search and selection makes all the difference (FDR>80%):
 - This is the same parametric model that the literature used—under the single-trial assumption—to justify FDR levels around 5%-15%



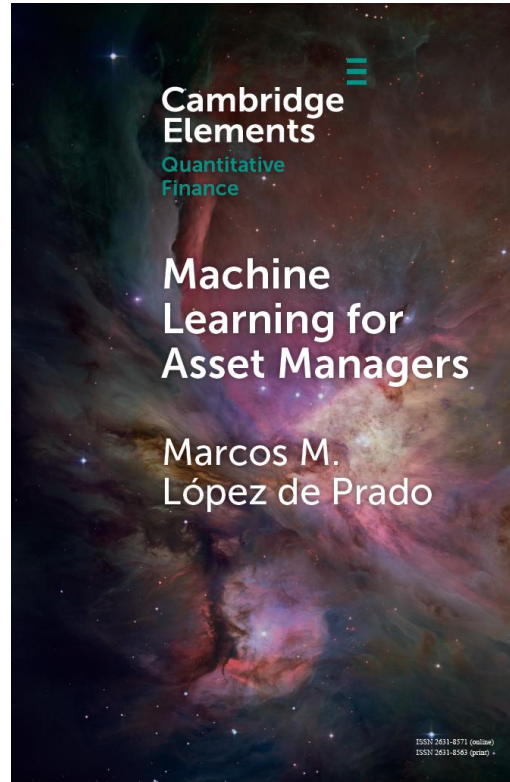
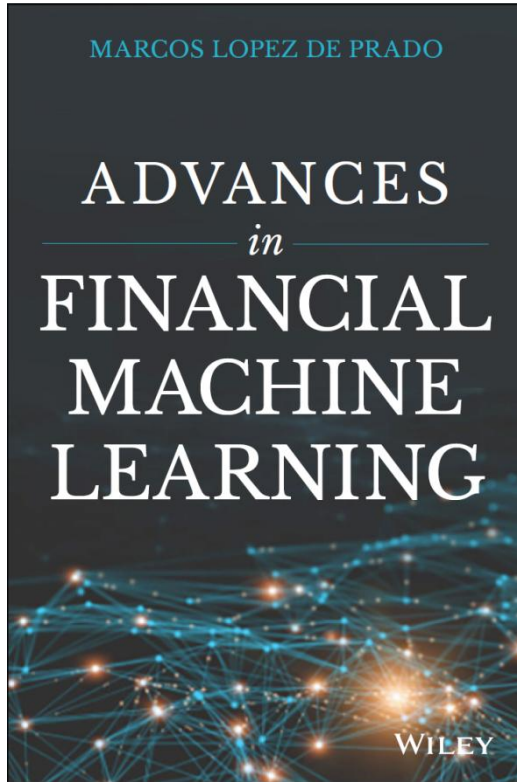
Weighted null and alternative densities implied by the fitted search-adjusted model at $K = 5$. This translates the empirical calibration into the same geometric decomposition used earlier in the numerical example, visualizing how the fitted model yields an FDR above 80%.

Conclusions

Conclusions

- Our findings are grouped in two important points:
 1. **Identification failure**: earlier attempts in the literature to estimate the FDR from the cross-section of reported in-sample statistics are misguided, because the FDR is not identifiable without making explicit assumptions about the latent search-and-selection process
 2. **Search-adjusted FDR estimation**: with explicit modelling of the search-and-selection process, fitting the parametric model used by earlier studies to the cross-section of reported in-sample statistics yields higher levels of FDR than previously reported
- **Corollary: The “factor zoo” should no longer be treated as a reliable map of investment opportunity**

For Additional Details



The first wave of quantitative innovation in finance was led by Markowitz optimization. Machine Learning is the second wave and it will touch every aspect of finance. López de Prado's Advances in Financial Machine Learning is essential for readers who want to be ahead of the technology rather than being replaced by it.
— Prof. **Campbell Harvey**, Duke University.
Former President of the American Finance Association.

Financial problems require very distinct machine learning solutions. Dr. López de Prado's book is the first one to characterize what makes standard machine learning tools fail when applied to the field of finance, and the first one to provide practical solutions to unique challenges faced by asset managers. Everyone who wants to understand the future of finance should read this book.

— Prof. **Frank Fabozzi**, EDHEC Business School.
Editor of The Journal of Portfolio Management.

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